**DO NOW — RECALL**

# Lesson 049 · Stack<T>: LIFO and Program State

Name: _____ Class: _____ Date: _____

**1. How does List<T>.RemoveAt differ from Remove?**

**2. What is a stack's removal order?**

**3. What is an instance method?**

**4. An empty stack executes push(A), push(B), push(C), x=pop(), push(D), push(x), then pop(). Trace the final stack from bottom to top.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

By index versus by matching value.

### 2. Model answer

LIFO.

### 3. Model answer

Method called on a specific object.

### 4. Model answer

A, B, D. x becomes C; pushing x restores C to the top, and the final pop removes C.

### By the end, you can…

• A stack is last-in, first-out (LIFO). Push places a value on top, Pop removes the top, and Peek reads without removing.

• A stack can support undo histories and recursive call frames. The physical call stack is a related example but not this Stack<T> collection.

• Always check Count or otherwise handle an empty-stack exception before Pop/Peek.

### Vocabulary — know these terms

LIFO — Last In First Out.

Push — Add to top of stack.

Pop — Remove and return top.

Peek — View top without removing.

NEW LEARNING

# Read, annotate and explain

### Observe

A stack is last-in, first-out (LIFO). Push places a value on top, Pop removes the top, and Peek reads without removing.

### Explain

A stack can support undo histories and recursive call frames. The physical call stack is a related example but not this Stack<T> collection.

### Apply

Always check Count or otherwise handle an empty-stack exception before Pop/Peek.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace stack push RED, push BLUE, pop.

### Trace and justify

Popped BLUE; remaining stack top RED. Most recent value leaves first.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Push A, B, C; Pop once, then Peek. State both outputs and remaining stack. [6]

### Question 2

Write C# to push 3,6,9 and safely pop one value. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why stack suits undo history but queue suits printing jobs; include two operations. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Peek permanently removes the last value from a stack.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.